

George Salib

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Publications & Projects

N-Body System Simulation Framework

July 2024

A **Lagrangian** framework to simulate Newtonian binary and multi-star systems using **Eulerian integration**. **Validated** with real data & **produced accurate results**.

Tech: Python, matplotlib.pyplot, matplotlib.animation, numpy, argparse

Celestial Classifier Trained on SDSS Data using Random Forest

April 2026

Implemented a **Random Forest** machine learning model to classify **100K** stars, galaxies, and quasars using **SDSS DR17** photometric data, achieving a **0.98 weighted F1-score**.

Tech: Python, scikit-learn, pandas, numpy, seaborn, matplotlib

A Comparison: Galactic Formation & Evolution through Semi-Analytic and Numerical Hydrodynamic Frameworks in the Context of Extragalactic Astronomy,
Youth Science Journal (supervised by the Egyptian Ministry of Education)

Oct. 2023

Multithreaded HTTP Web Server

November 2025

A **multithreaded** web server capable of serving static content. Handles standard GET requests & returns proper error responses using a **thread pool executor**.

Tech: Python, socket, concurrent.futures, os, sys

Experience

Lead Organizer — Scrapyard Giza, sponsored by the Hack Foundation

Mar. 2025

Organized **Egypt's largest high school hackathon** with **45 projects** and **150 attendees**. Managed part of logistics, finances, and day-of operation. **Raised \$2,000+**. Event attracted **475+ signups**, ranking **#1 among 100+ cities** globally

Lead Organizer, Logistical & Financial Administrator — Counterspell Giza, sponsored by ChuckleFish Limited & Swish Swoosh

Nov. 2024

Organized a 12-hour hackathon with **30+ projects** and **100 attendees**. **Fundraised \$1,500** and prizes from sponsors. Managed day-of operation budgeting, logistics, and financials including venue, t-shirts, and catering. Negotiated a 5% discount with food supplier and Increased budget with partner. Event attracted **280+ sign-ups (#1 of 100+ cities** globally)

Vice President — Hack Club, an international non-profit, STEM Egypt Branch

Jan. 2024 – Feb. 2025

Oversaw **9 mentors**, **90+ sessions**, and **70+ members**. Organized a 4-week boot camp, consisting of **5 CS tracks** and **125+ mentees** and Handpicked its mentors. Supervised a CP competition with **225+ contestants from 20 countries**

Cyber Security Mentor — STEM Egypt Hack Club

Jan. 2024 – Feb. 2025

Taught **20 students** in **14+ weekly sessions** (2024). Mentored **4 students** in an **intensive four-week boot camp** (2023). Prepared curriculum and tasks and Evaluated submissions during both positions

Core Competencies

Programming: **Python** (Proficient); **C++** (Competent); **Bash** (Intermediate); **MySQL** (Basic Familiarity); **LaTeX** (Intermediate)

Tech Used: **Git**, **Raspberry Pi**, **Linux** (Ubuntu, RPiOS & Kali), **VirtualBox**, **Wireshark**, **Nmap**, **Burp Suite**, **EasyEDA**, **Arduino**, **Blender**

Languages: **English** (Fluent), **Arabic** (Native), **French** (Limited Working Proficiency)

Honors, Awards & Certifications

International Astronomy and Astrophysics Competition — **Top 3% of ~5040 competitors**, Gold Honor;
Top 20% of 4500+, Silver Honor

2024 & 2023

Immerse Education Essay Competition — **Top 7–8% of ~6000 entrants**, awarded ~\$1000 scholarship for a 550-word research essay on stellar evolution

Jan. 2024

PHYS-105-623 (HVCC) — **Received ~\$630 merit scholarship** to take a college-level physics course at Hudson Valley Community College

July 2024

EducationUSA Connects+ Program — Received ~\$900 scholarship funded by the U.S. Department of State to attend a summer course at St. Cloud State University	July 2023
STEM Schools Entrance Exam — Top 0.6% of 20,000 students	Aug. 2022
DET, CEFR:C1 — 150 Literacy, 150 Comprehension, 140 Production, 135 Conversation	Jan. 2025
ACT, 34 Composite — 36 Science, 34 English, 33 Maths, 32 Reading	Sept. 2024

Education	
STEM High School for Boys, 6th of October Coursework: 4.0 GPA (G10, G11, G12); Mathematics (Adjacent to AP), Mechanics (Adjacent to AP), Chemistry (Adjacent to AP), Physics (Adjacent to AP), Robotics (Elective), Electronics (Elective)	Giza, Egypt
Hudson Valley Community College (Virtual) Course Identifier: Introduction to Astronomy (PHYS-105-623) Coursework: Earned 3 college credits (4.0 GPA); Engaged in astronomical calculations and simulations in six labs, and collaborative discussions	NY, U.S.
University of Helsinki (Virtual) Course Identifier: Elements of AI (An Introduction to AI) Coursework: Earned 2 ECTS credits; Explored core concepts in artificial intelligence, including machine learning, neural networks, GANs, and societal implications	Helsinki, Finland
Saint Cloud State University (Virtual) Course Identifier: Academic English and Composition (EAP 195-55) Coursework: Earned 1 college credit (4.0 GPA); Developed my unique writing style and honed my academic writing. Had discussions and debates with peers from around the globe	MN, U.S.